

12 A kinetic energy is directly proportional to thermodynamic temperature

$$\frac{3}{2}kT = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3k}{m}}(\sqrt{T})$$

$$\frac{v_{\text{new}} - v_{\text{old}}}{v_{\text{old}}} \times 100\%$$

$$= \frac{\sqrt{T_{\text{new}}} - \sqrt{T_{\text{old}}}}{\sqrt{T_{\text{old}}}} \times 100\%$$

$$= \frac{\sqrt{273.15 + 40.5} - \sqrt{273.15 + 32.1}}{\sqrt{273.15 + 32.1}}$$

$$= 1.4\%$$

13 D $pV = NkT \rightarrow p = \frac{Nk}{V}(T)$